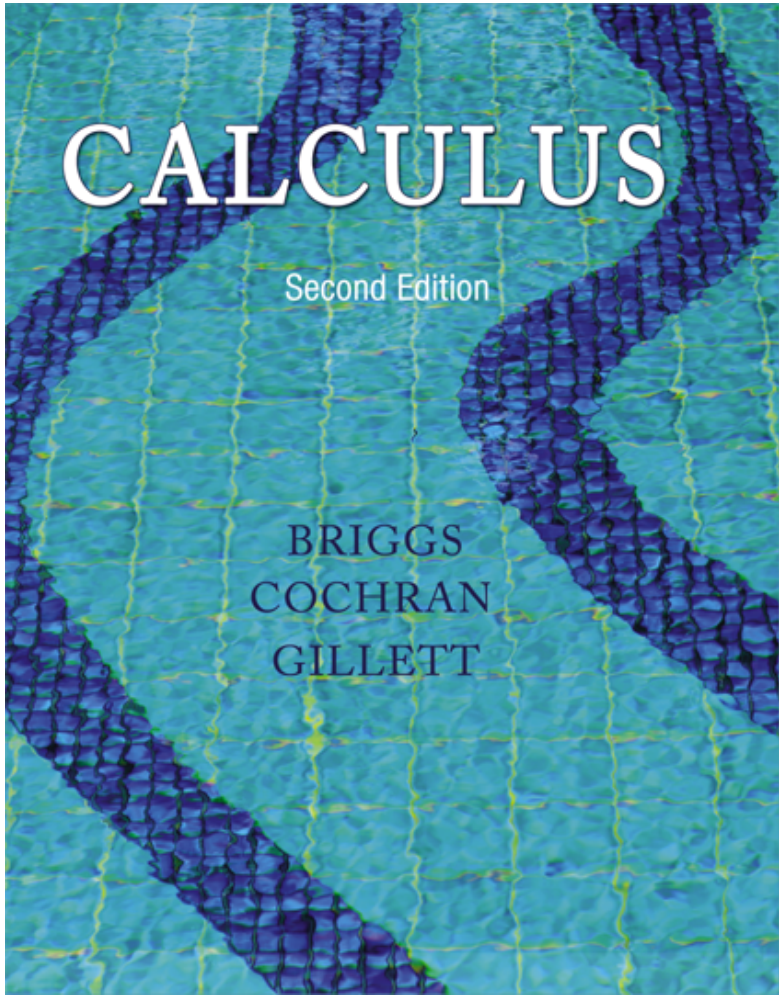
Chapter 2

Limits



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2.6

Continuity

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Figure 2.42 (a)

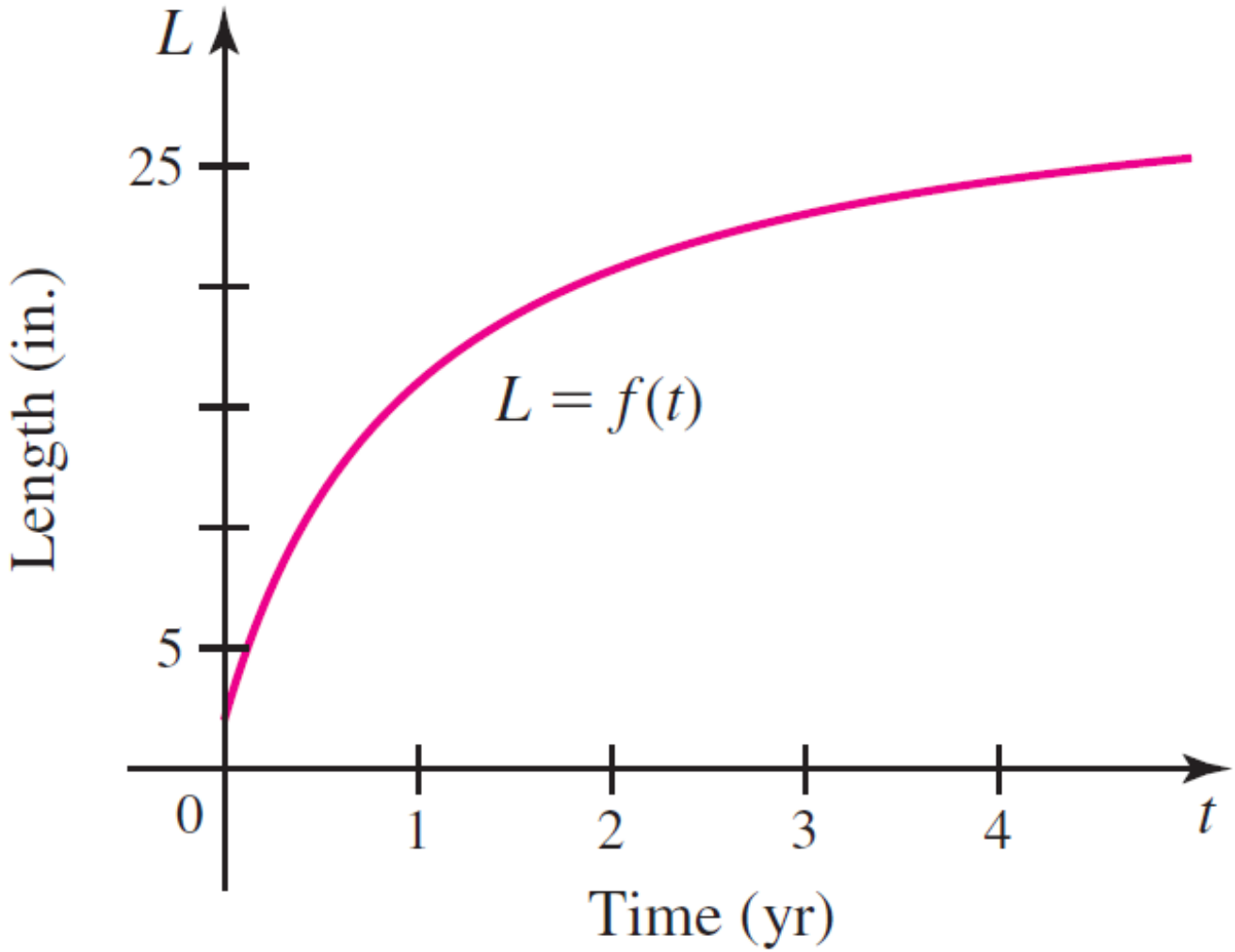

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Figure 2.42 (b)

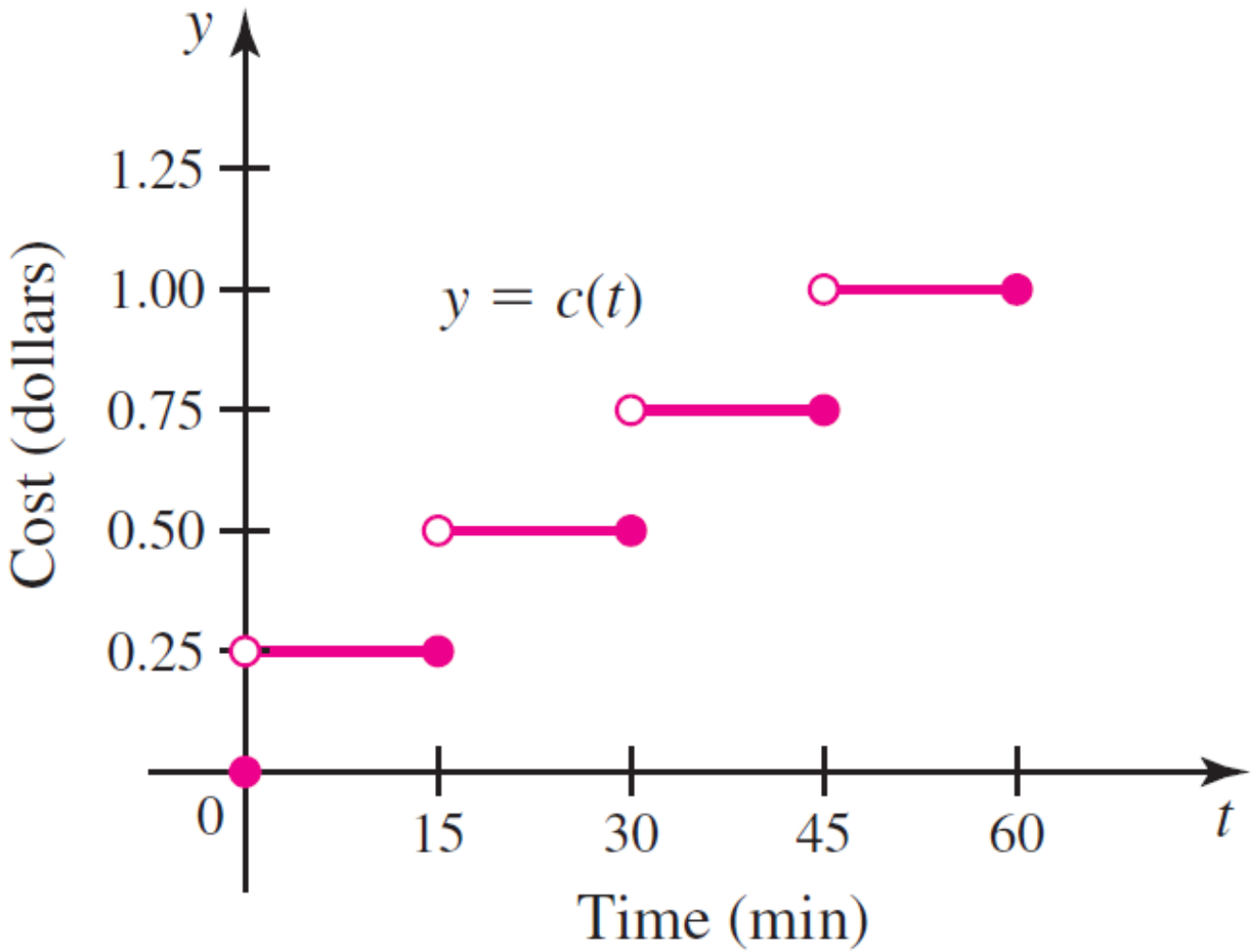

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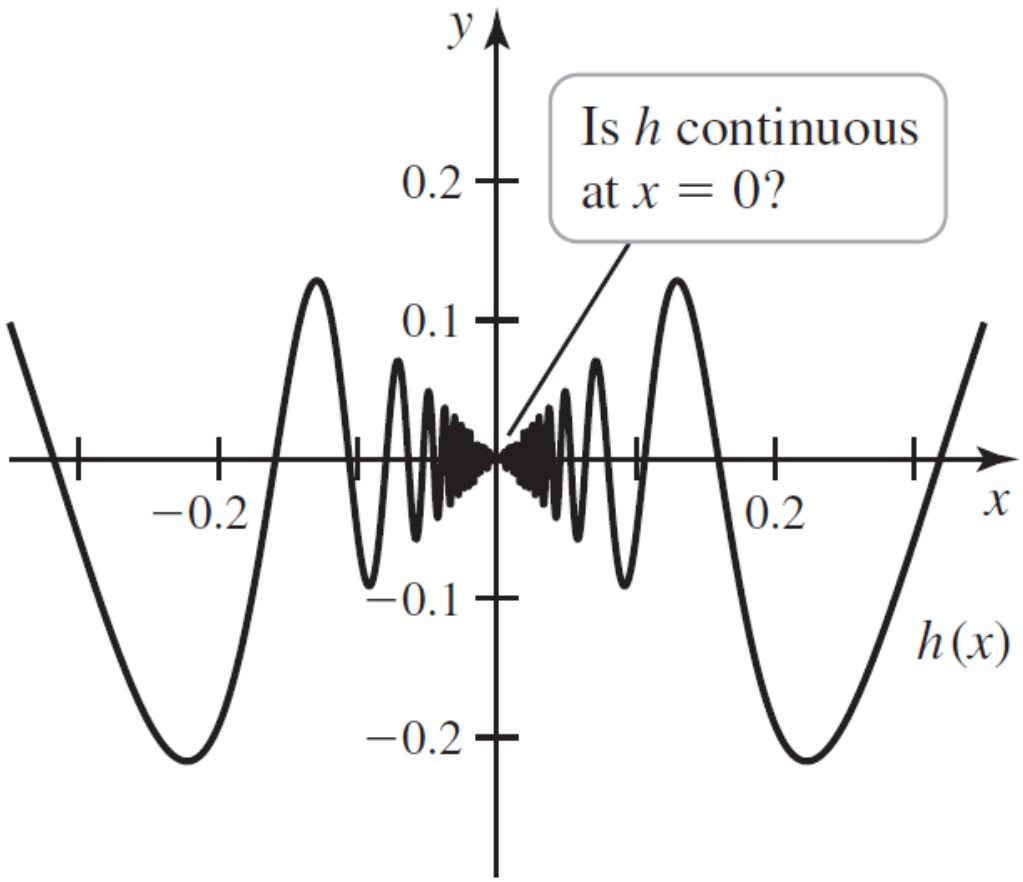
Figure 2.43



$$h(x) = \begin{cases} x \sin \dfrac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

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DEFINITION Continuity at a Point

A function f is **continuous** at a if $\lim_{x \to a} f(x) = f(a)$. If f is not continuous at a, then a is a **point of discontinuity**.

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Continuity Checklist

In order for f to be continuous at a, the following three conditions must hold:

1. $f(a)$ is defined (a is in the domain of f).
2. $\lim_{x \to a} f(x)$ exists.
3. $\lim_{x \to a} f(x) = f(a)$ (the value of f equals the limit of f at a).

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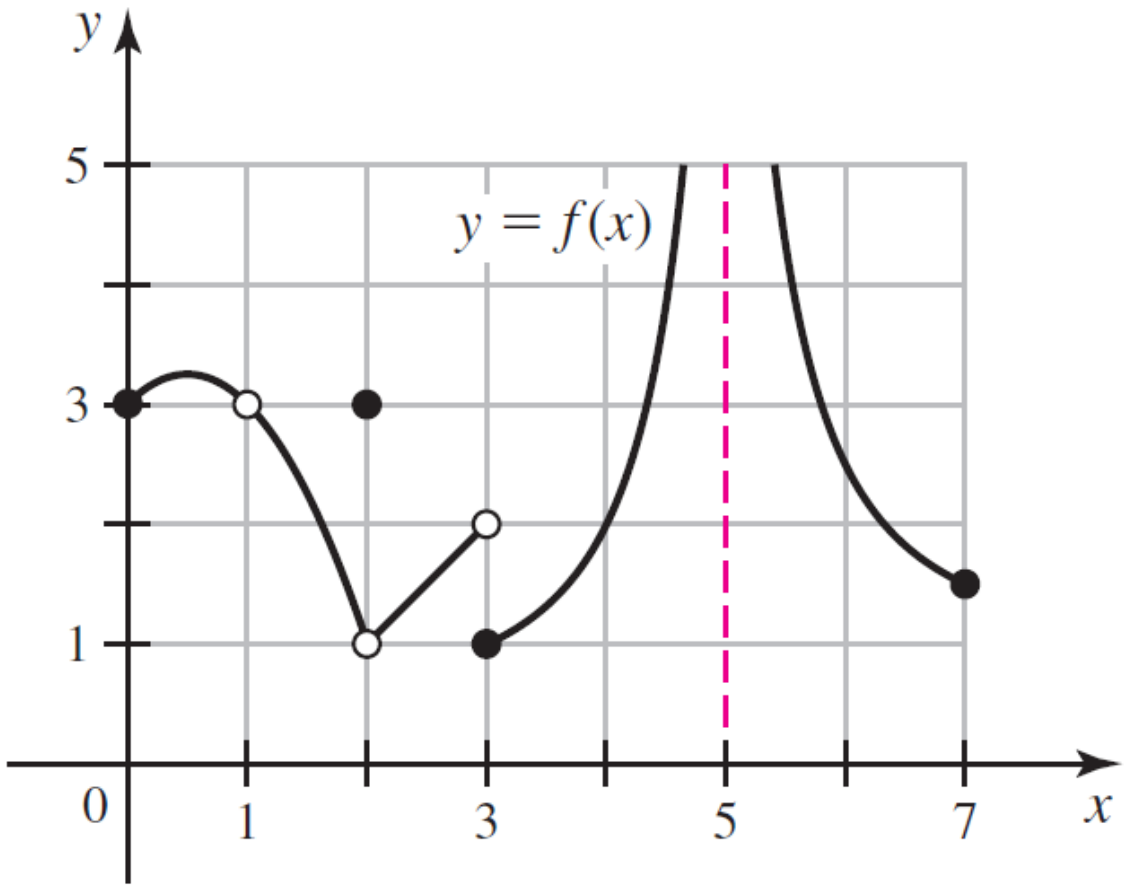
Figure 2.44



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THEOREM 2.8 Continuity Rules

If f and g are continuous at a, then the following functions are also continuous at a. Assume c is a constant and $n > 0$ is an integer.

a. $f + g$

b. $f - g$

c. cf

d. fg

e. f/g, provided $g(a) \neq 0$

f. $(f(x))^n$

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THEOREM 2.9 Polynomial and Rational Functions

a. A polynomial function is continuous for all x.

b. A rational function (a function of the form $\frac{p}{q}$, where p and q are polynomials) is continuous for all x for which $q(x) \neq 0$.

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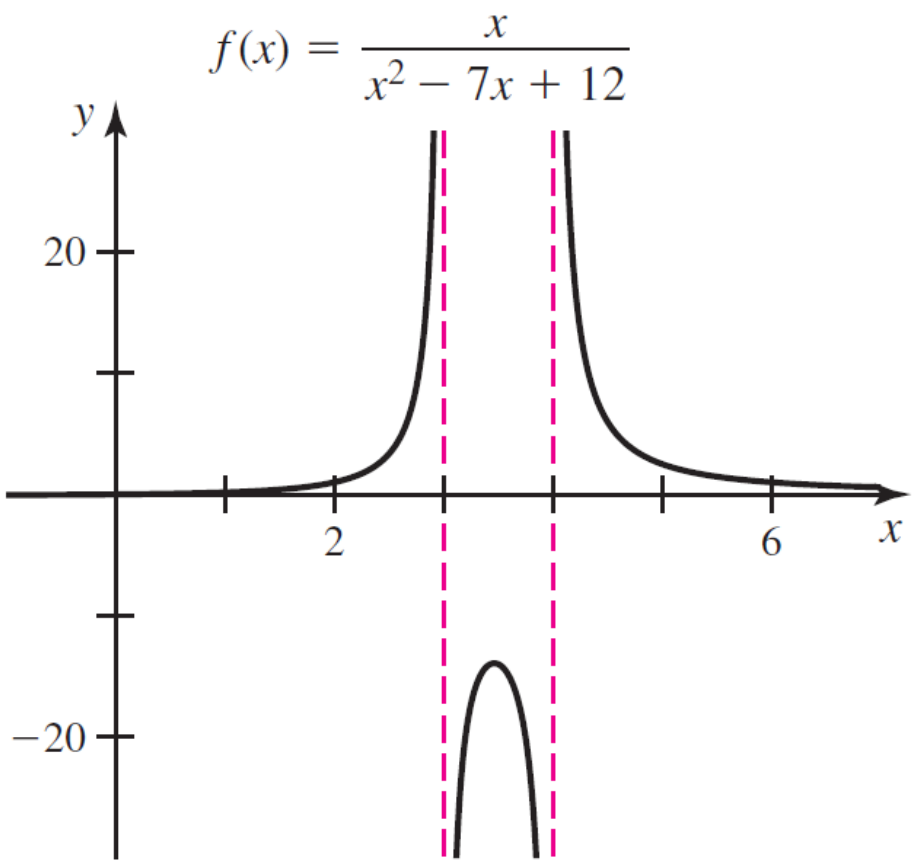
Figure 2.45



Continuous everywhere except $x = 3$ and $x = 4$

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THEOREM 2.10 Continuity of Composite Functions at a Point

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ is continuous at a.

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THEOREM 2.11 Limits of Composite Functions

1. If g is continuous at a and f is continuous at $g(a)$, then

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right).$$

2. If $\lim_{x \to a} g(x) = L$ and f is continuous at L, then

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right).$$

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Figure 2.46 (a)

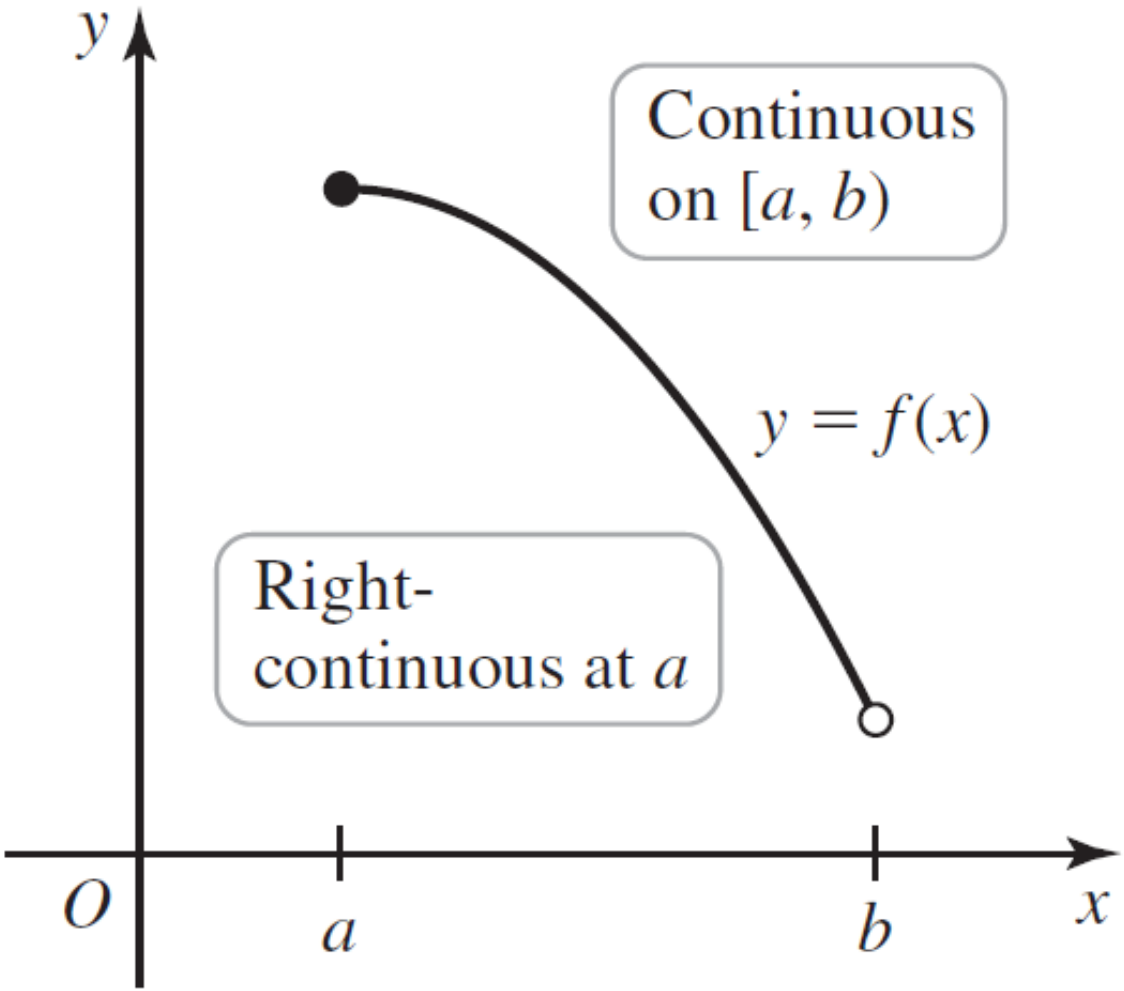

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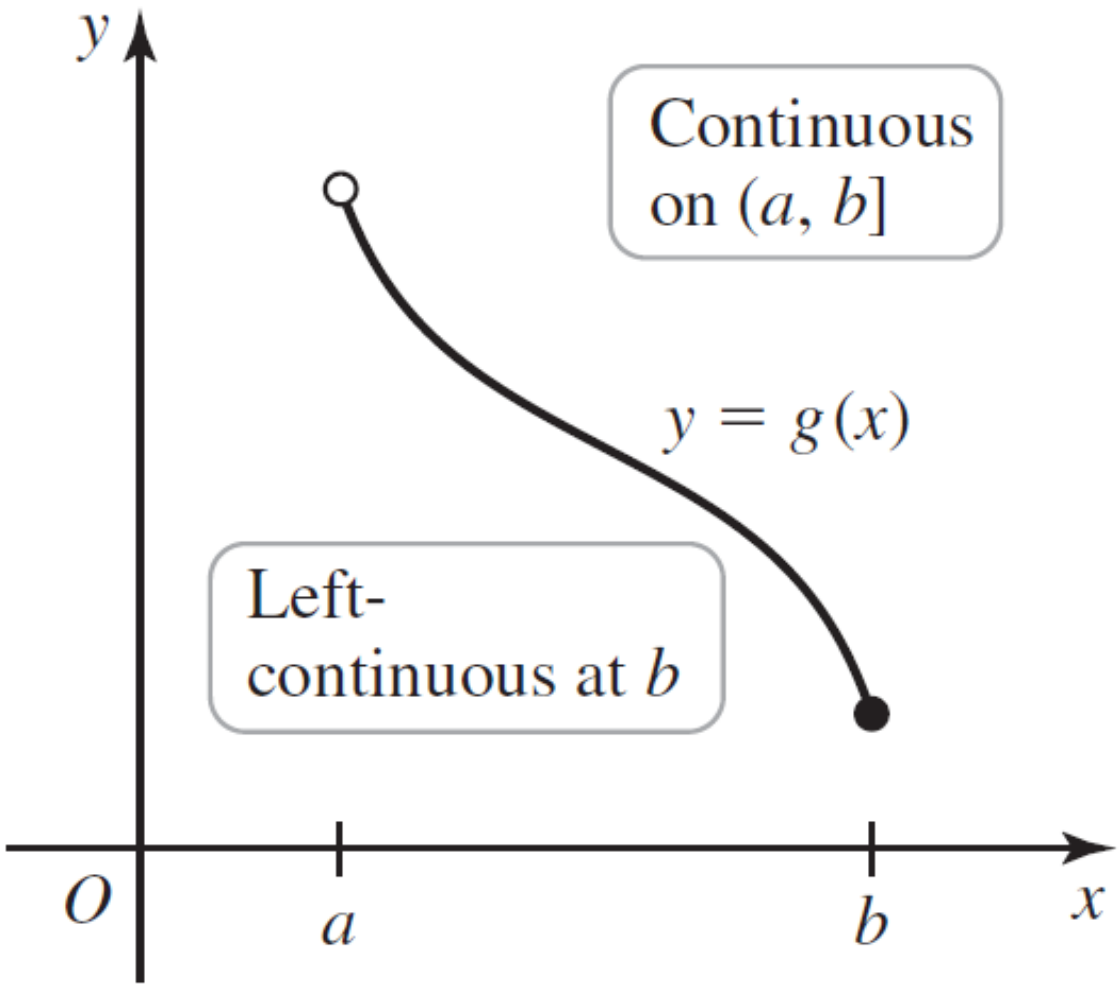
Figure 2.46 (b)



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DEFINITION Continuity at Endpoints

A function f is **continuous from the right** (or **right-continuous**) at a if $\lim_{x \to a^+} f(x) = f(a)$ and f is **continuous from the left** (or **left-continuous**) at b if $\lim_{x \to b^-} f(x) = f(b)$.

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DEFINITION Continuity on an Interval

A function f is **continuous on an interval I** if it is continuous at all points of I. If I contains its endpoints, continuity on I means continuous from the right or left at the endpoints.

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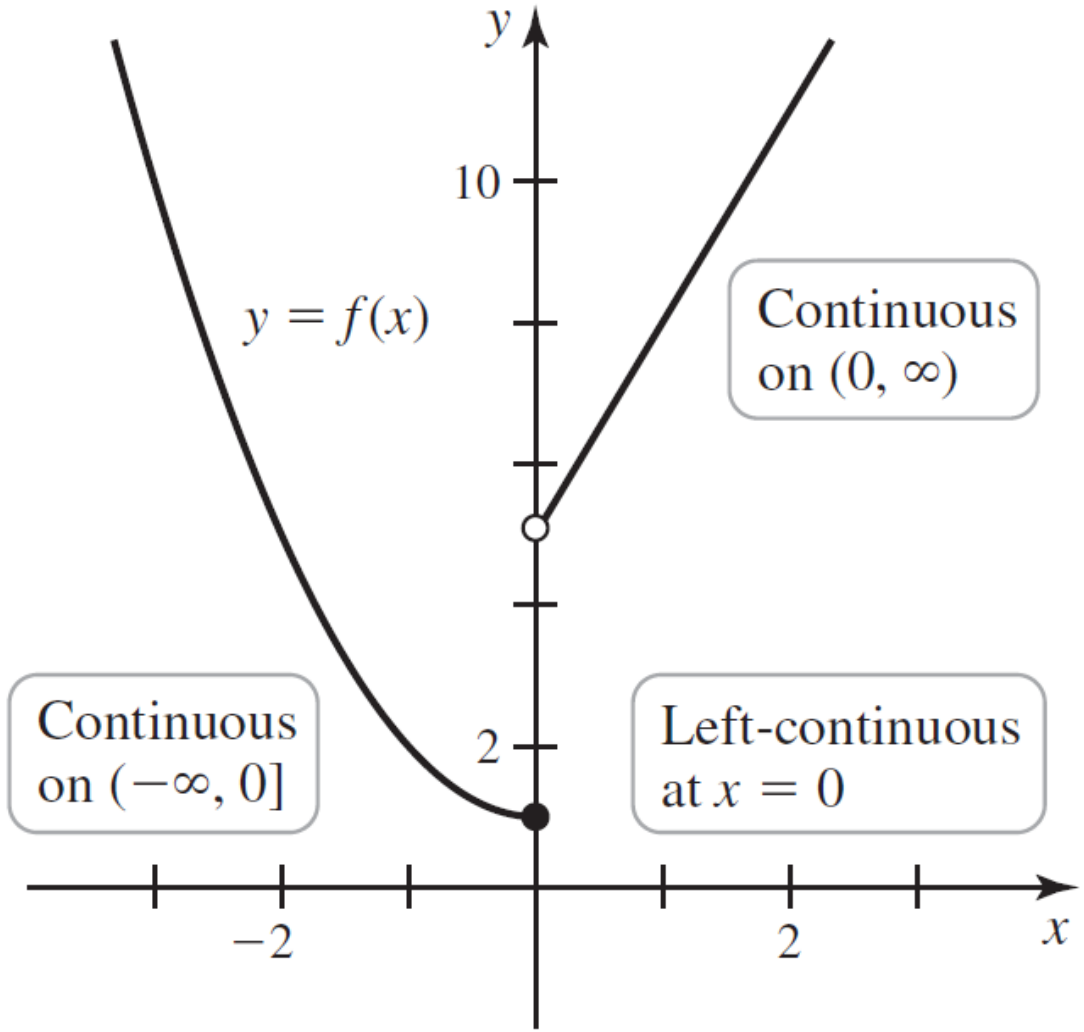
Figure 2.47



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THEOREM 2.12 Continuity of Functions with Roots

Assume that m and n are positive integers with no common factors. If m is an odd integer, then $(f(x))^{n/m}$ is continuous at all points at which f is continuous.

If m is even, then $(f(x))^{n/m}$ is continuous at all points a at which f is continuous and $f(a) > 0$.

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Figure 2.48

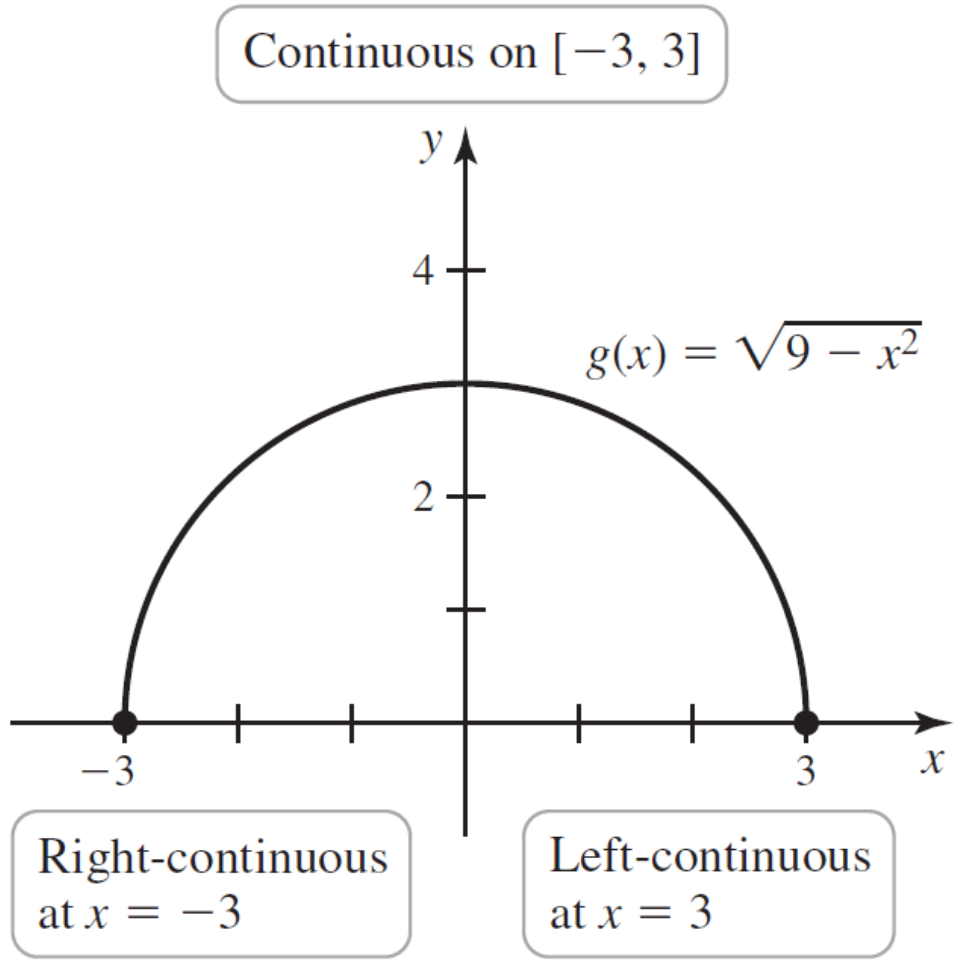

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Figure 2.49

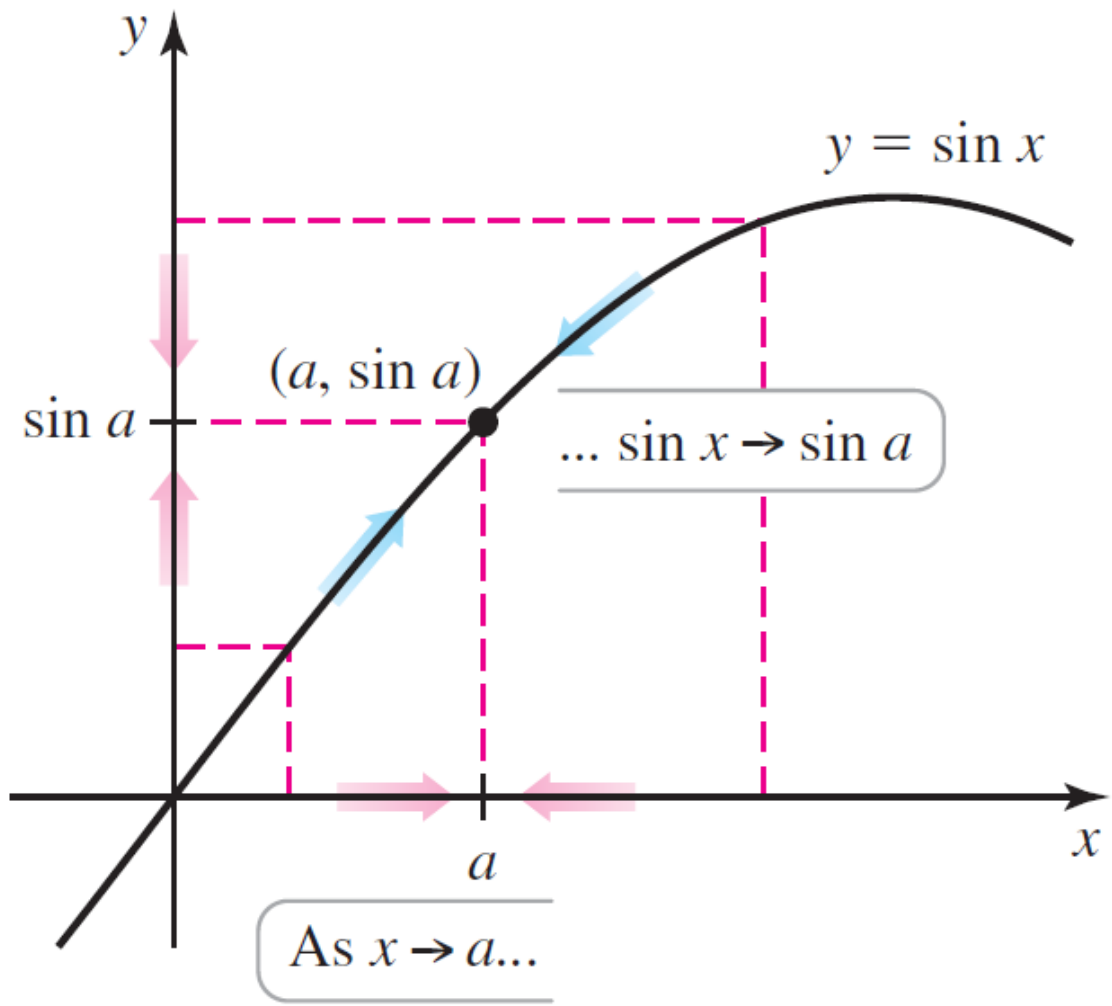

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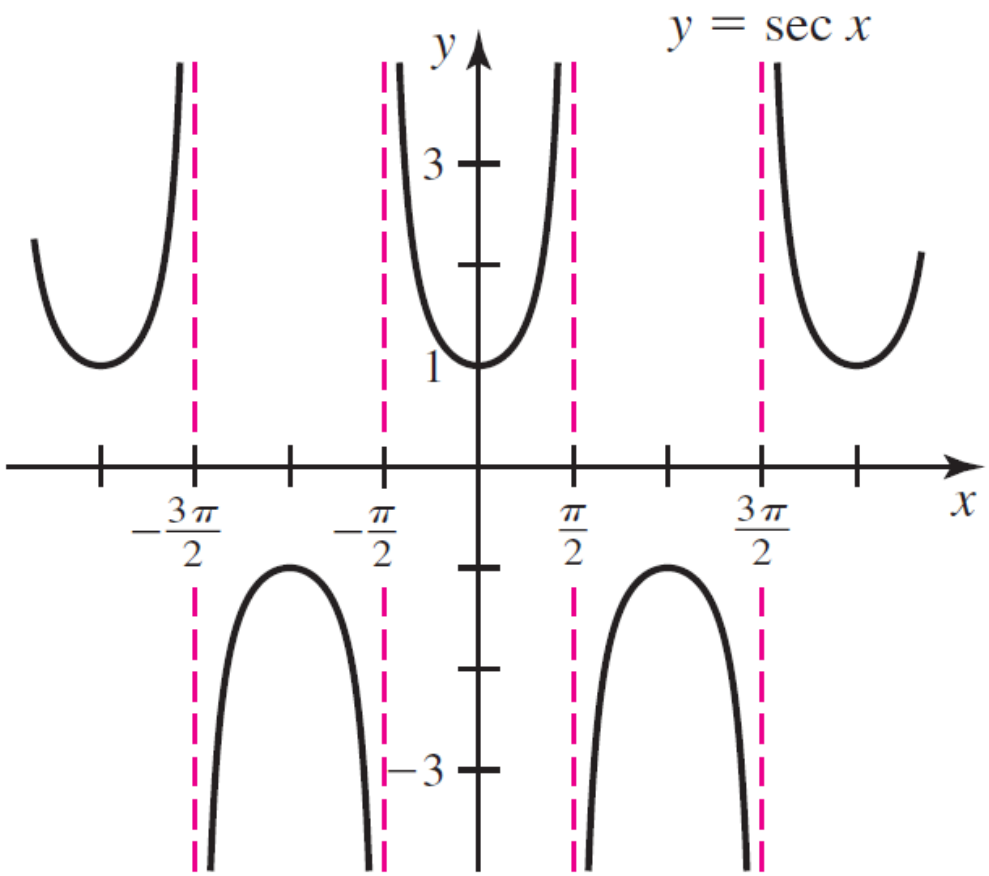
Figure 2.50



sec x is continuous at all points of its domain.

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THEOREM 2.13 Continuity of Trigonometric Functions

The functions $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$ are continuous at all points of their domains.

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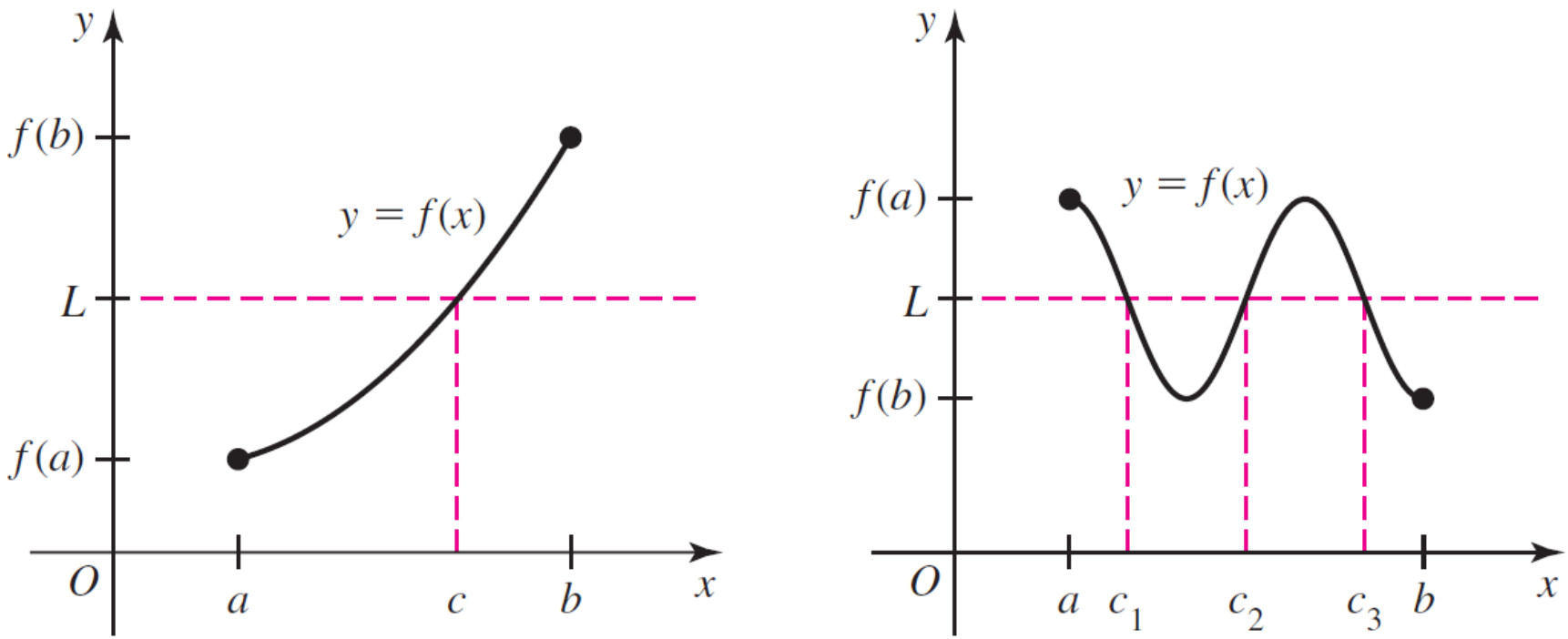
Figure 2.51

Intermediate Value Theorem



In (a, b) there is *at least* one number c such that $f(c) = L$, where L is between $f(a)$ and $f(b)$.

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THEOREM 2.14 Intermediate Value Theorem

Suppose f is continuous on the interval $[a, b]$ and L is a number strictly between $f(a)$ and $f(b)$. Then there exists at least one number c in (a, b) satisfying $f(c) = L$.

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Figure 2.52

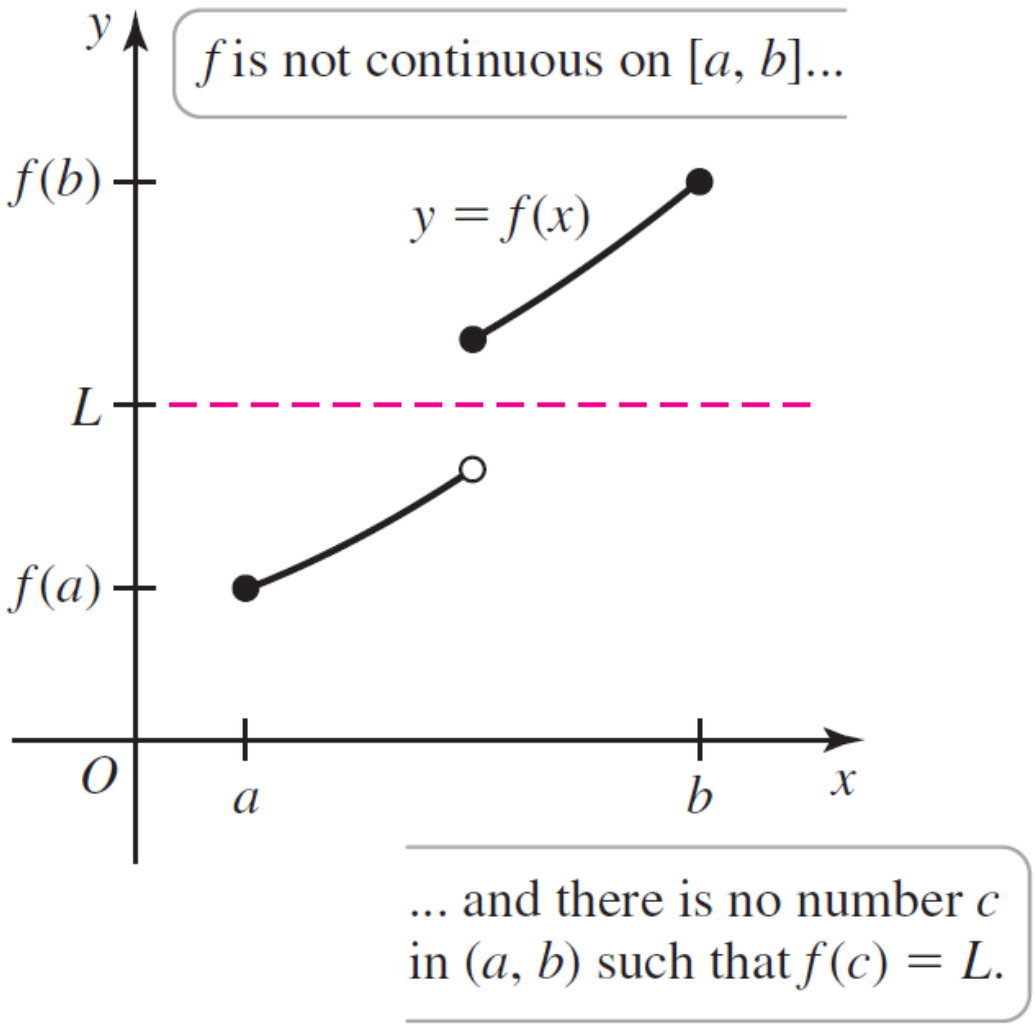

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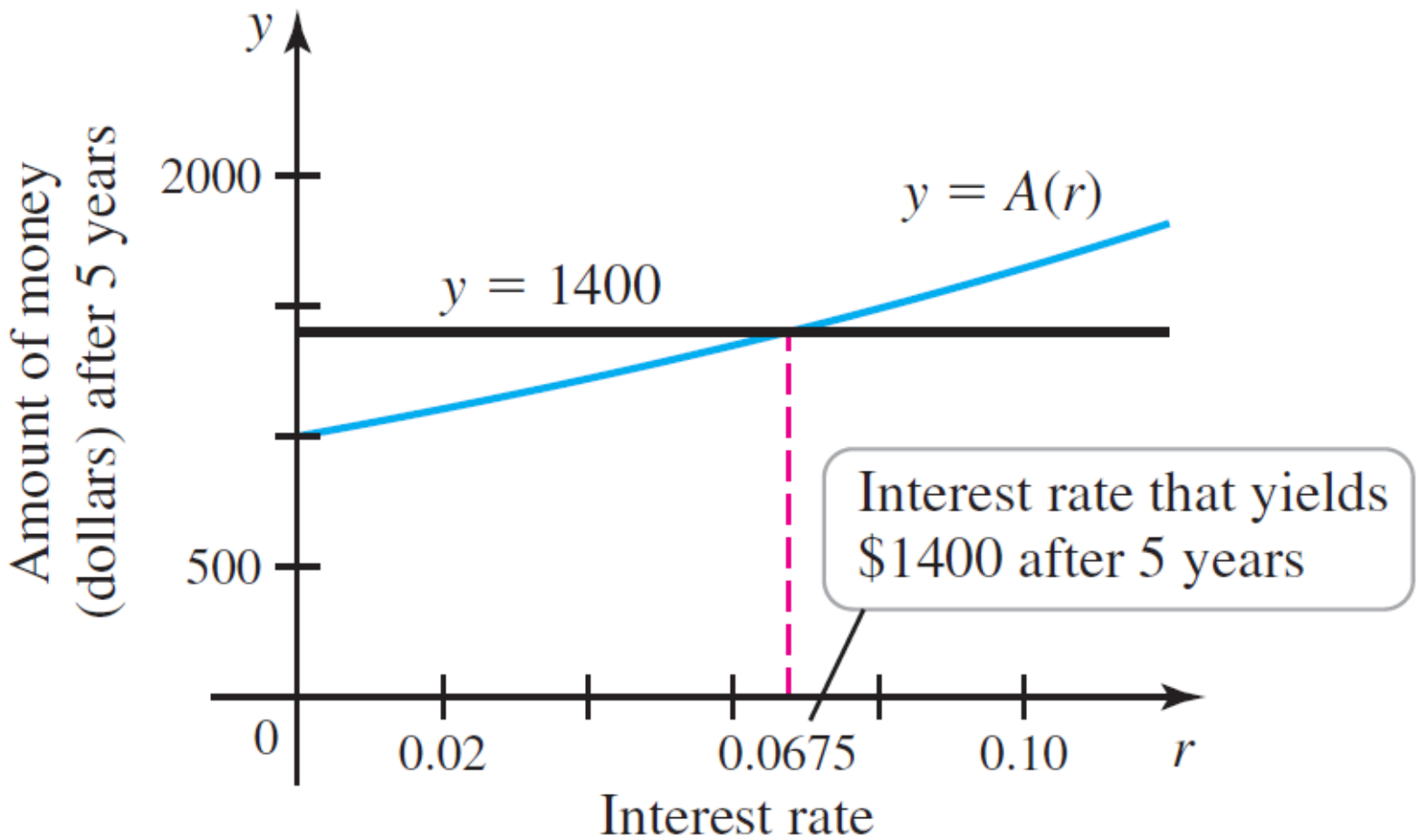
Figure 2.53



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